Kexuan Zhang

PhD Researcher · Graphics and Systems Programmer
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Education and Research

University of Leeds, UK 2025 – Present

PhD in Perceptual Graphics
Research: virtual reality, foveated rendering and high-performance graphics.
Focus: stereo reprojection and image inpainting.

University of Leeds, UK 2023 – 2024

MSc in High-Performance Graphics and Games Engineering, Distinction
Average: 74.7. Dissertation: real-time global illumination with Light Propagation Volumes.

Jianghan University, Wuhan, China 2017 – 2021

BEng in Computer Science and Technology (Software Engineering)
Average: 84.3/100; rank: 18/212.

Software Engineering Experience

Shanghai Zihao Information Technology Co., Ltd Jul. 2022 – May 2024

Game Server Development Engineer

- Developed high-performance .NET game server systems and investigated performance bottlenecks and MongoDB integration.
- Built Unity skill-pipeline tools, Excel configuration tooling and Jenkins CI/CD pipelines.
- Developed Roslyn-based static analysis tools to enforce code consistency.

Shanghai Zihao Information Technology Co., Ltd Dec. 2021 – Jun. 2022

Intern Programmer

- Built experience with C#, Win32, multithreading, networking and Unix/Linux systems.

Selected Projects

VRI – Cross-platform Render Hardware Interface Active development

C ABI, C++23, Vulkan, D3D12, Metal, WebGPU, OpenGL Personal project

- Develop an explicit rendering API with command buffers, descriptor sets and synchronization, paired with a header-only C++23 RAII wrapper.
- Implement backend capability queries, validation and portable Slang shaders. Browser examples and a Vulkan path tracer demonstrate the API; feature coverage is documented per backend.

Light Propagation Volumes 2024

C++, OpenGL, Real-time Global Illumination MSc dissertation

- Implemented an LPV renderer on VGFW to explore real-time indirect lighting.
- Integrated cascaded shadow maps, HBAO, screen-space reflections, FXAA, an ACES filmic tone-mapping approximation and bloom into the rendering pipeline.

libvultra – Graphics Research Framework 2024 – Present

C++23, Vulkan, Slang, Render Graphs Personal project

- Develop the current VRI-based framework for small rendering demos and research experiments on Windows and Linux.
- Build reusable rendering, shader and asset components; the early-stage development branch includes a glTF viewer, render-graph construction and shader compilation/reload.

Cells of Division – High Path Games 2024 – Present

Co-founder and Programming Lead; Godot Independent game

- Design the program architecture and core gameplay subsystems for a bullet hell roguelike; lead the programming team and coordinate development.
- Built framework systems including ECS, configuration, localisation and buffs. A free demo is available on Steam.

OmniLyrics – Cross-platform Lyrics Tool Active development

C#, .NET 10, Avalonia, CLI/TUI Personal project

- Built a shared .NET core for desktop, terminal and status-bar interfaces, with bilingual karaoke lyrics and platform playback integrations.
- Publish Windows, macOS and Linux releases, including signed and notarized macOS GUI applications.

Technical Skills

- Languages: C/C++, C#, Python, Java and Lua. Currently studying CUDA independently through CIS5650.
- Graphics and engines: Vulkan, OpenGL, Slang, render graphs, global illumination, Unity and Godot.
- Systems and tooling: multithreading, networking, .NET performance, Git, xmake/CMake, Jenkins, Roslyn and MongoDB.
- Languages spoken: Mandarin (native); English (IELTS 6.5).

Publication

Li, S. and **Zhang, K.** (2021). Mobile Surveillance System Based on Remote APP Control. *Computer Systems & Applications*, 30(06), 82–87. doi:10.15888/j.cnki.csa.007939. Second author.

Honors and Awards

National First Prize, Bochuang Cup AI Competition 2019, 2020
Provincial Second Prize, Blue Bridge Cup Programming Competition 2018, 2019
Provincial Third Prize, Programming Ladder Competition 2018, 2019